FINAL PROJECT DATA ANALYST (2026)

Final Project
Projet final

WRITER

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INDUSTRY

Fintech / P2P Lending & Paylater

Analyst Profile & Portfolio Overview



Nabil Ihsan

Data Analyst

CORE SKILL SETS

- Python (Pandas/Seaborn)
- SQL (PostgreSQL)
- Star Schema
- Tableau Public

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PROJECT HIGHLIGHT 01

TikTok Affiliate Data Analytics

Engineered a Star Schema in PostgreSQL and designed interactive Tableau dashboards to optimize Gross Merchandise Value (GMV) and affiliate campaign performance.

PROJECT HIGHLIGHT 02

HR Employee Attrition & Performance Analytics

Analyzed turnover drivers using Python and SQL to visualize HR retention risk levels and facilitate periodic performance reviews.

Final Project Focus & Business Scope

Mitigating P2P Lending & Paylater Default Risk in Indonesia: Portfolio Credit Risk Analysis via Early Warning System (EWS) Framework & Customer Risk Segmentation

10.000

Borrowers Data

Granular evaluation leveraging real-world credit and transaction logs.

0.80×

DSR Threshold

Mapping the safe-zone threshold for borrower debt service capacity.

0.06%

Optimal Default Target

Projected default rate reduction post-filtering implementation.

Portfolio Health Executive Breakdown

TOTAL OUTSTANDING

Rp99,01 M

Total exposure across 10,000 active P2P Lending & Paylater borrowers.

Default Rate (TWP90)

25.38%

Delinquency >90 days severely exceeding healthy tolerance baselines.

17.31% Special Mention Ratio
Vulnerable portfolio in DPD 1–30 position, prone to deterioration without EWS intervention.

1.68× Average DSR (Defaulted Pool)
Monthly debt obligations exceed clean monthly net income by 168%.

58.38% Behavioral Root Causes
Combination of Impulsive/FOMO Spending (35.88%) & Debt Spiral Trap (22.50%).

Projected Solution Impact
Default Rate drops down to 0.06% by enforcing DSR $\le$ 0.8× cap and maximum 3 active lending apps.

Market Background & Business Urgency

01

Industry Context

Rapid growth of digital financial inclusion (P2P Lending & Paylater) directly correlates with escalating non-performing loan exposure in Indonesia.

Rapid Penetration

02

Business Urgency

High non-performing loans erode capital recovery rates, hurt lenders, and risk breaching regulatory TWP90 thresholds set by OJK.

Capital & Reputational Risk

03

Core Objective

Build high-precision credit scoring filters and establish an Early Warning System (EWS) to flag high-risk borrowers prior to write-off.

Early Warning System

Problem Statement & Framework EDA

PROBLEM STATEMENT

Elevated default rate (25.38%) severely threatens liquidity and overall fintech portfolio health.



Q1 — Collectibility & Sector

How is default risk distributed across employment sectors?

Q2 — Financial Ratios

What critical DSR and limit utilization thresholds trigger borrower default?

Q3 — Behavioral Drivers

What primary psychological/economic root causes drive default?

Q4 — Strategic Mitigation

How can we design a DSR-based Early Warning System to suppress Default Rate?

Data Understanding & Preprocessing Workflow

Phase 1 (Raw Dataset)

10,000 Rows

- **26 Original Columns** (0 missing values)
- **Demographics** (Job, Province, Age)
- **Financials** (Income, Limit, Tenor, Installment)
- **Status** (DPD, Collectibility).

Phase 2

Feature Engineering

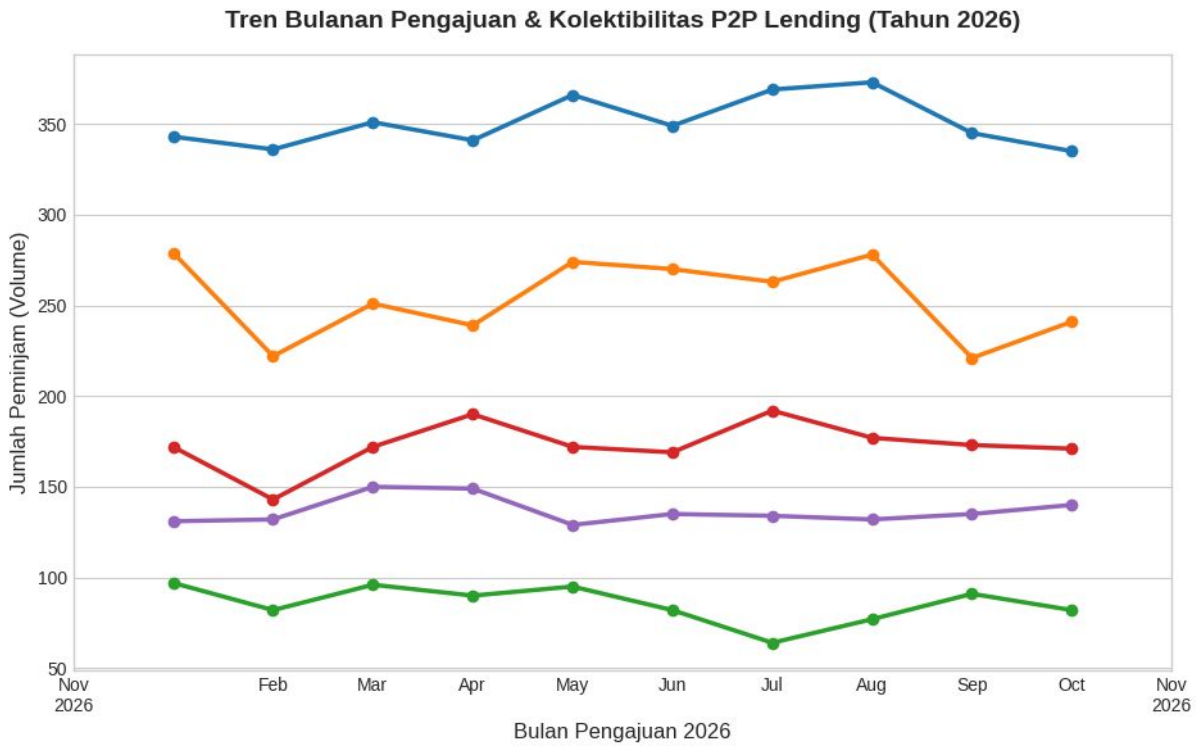
- **Calculated DSR** (Installment / Net Income)
- **Limit Utilization** (Outstanding / Limit)
- **Distress Index** (Categorical risk tagging)
- **Class Mapping** (TWP90 vs Performing grouping).

Phase 3

27 Refined Features

- 100% consistent dataset free of extreme outliers
- Standardized quantitative & categorical features ready for EDA and segmentation modeling.

Monthly Trend Fluctuations Indicate Consistent Systemic Risk Throughout 2026



1. Default Volume Stability

TWP90 default lines (orange) and Current-Performing lines (blue) move in parallel monthly, indicating default risk is driven by persistent exposure rather than seasonal anomalies.

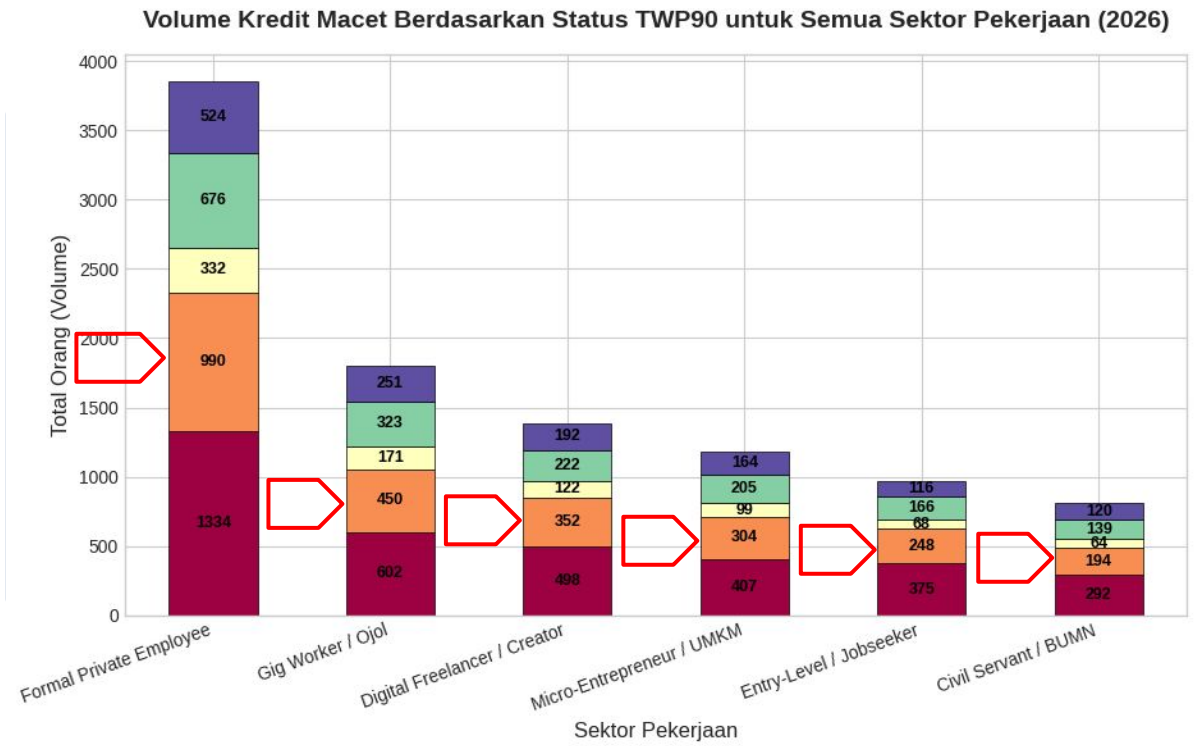
2. Mid-Year Risk Spikes

Volume spikes in Special Mention (red) appear around mid-year (April & July), serving as early warnings before borrowers plunge into total default.

3. Business Implication

Credit scoring policies and EWS monitoring must operate in real-time year-round rather than reactively.

Default Volume Heavy Concentration in Formal & Gig Worker Sectors



1. Highest Nominal Loss Pocket

Formal Private Employees (990 defaulted borrowers) and Gig Workers / Ojol (450 defaulted borrowers) dominate total portfolio risk exposure.

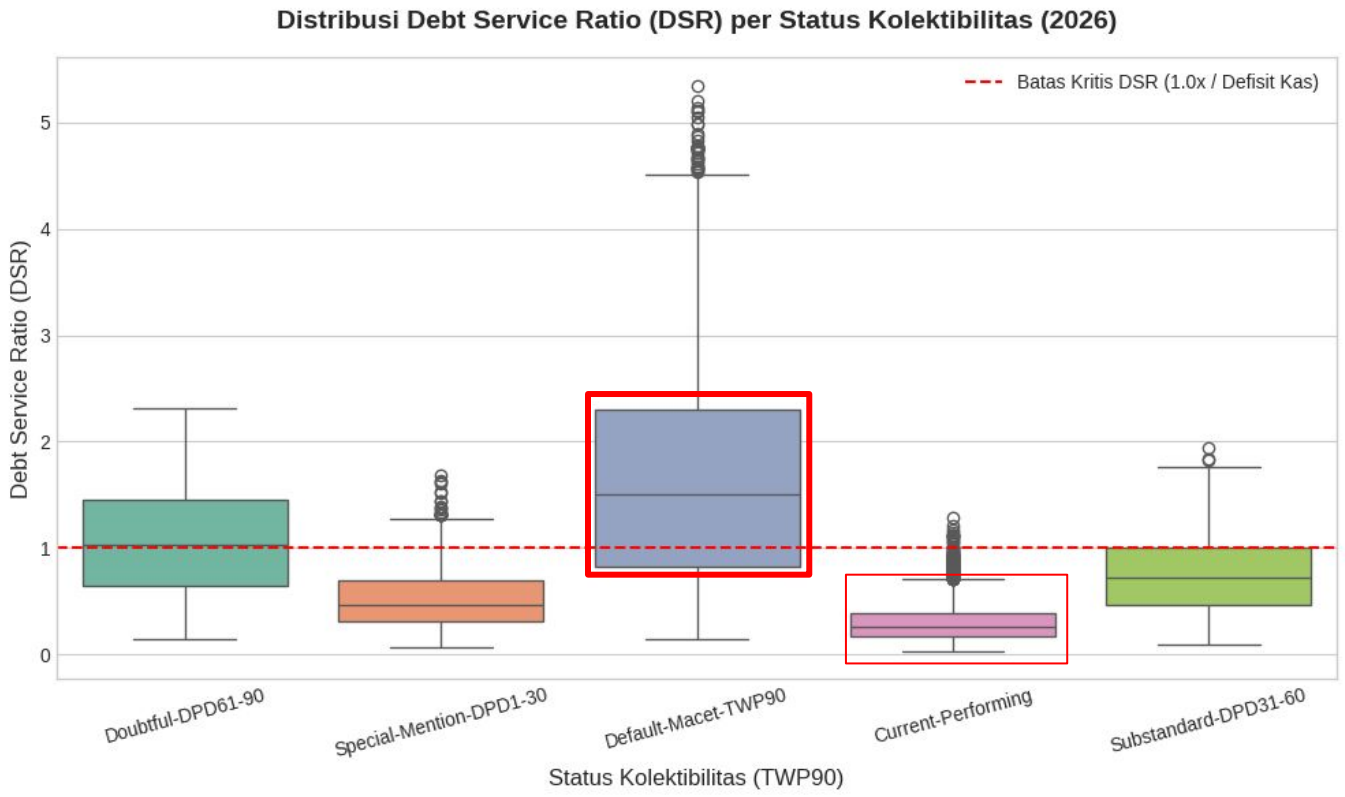
2. Formal Sector Risk Accumulation

Although default percentages look even (~25%), formal and gig workers carry the largest absolute default burden due to multi-app borrowing.

3. Business Implication

Limit caps and credit scoring tightening must aggressively prioritize these two primary segments to safeguard cash flows.

DSR > 80% & Limit Utilization 87% Represent the Point of No Return



1. Explosive Financial Ratio Contrast

Defaulted borrowers carry an average installment load of 1.68× their total clean monthly net income.

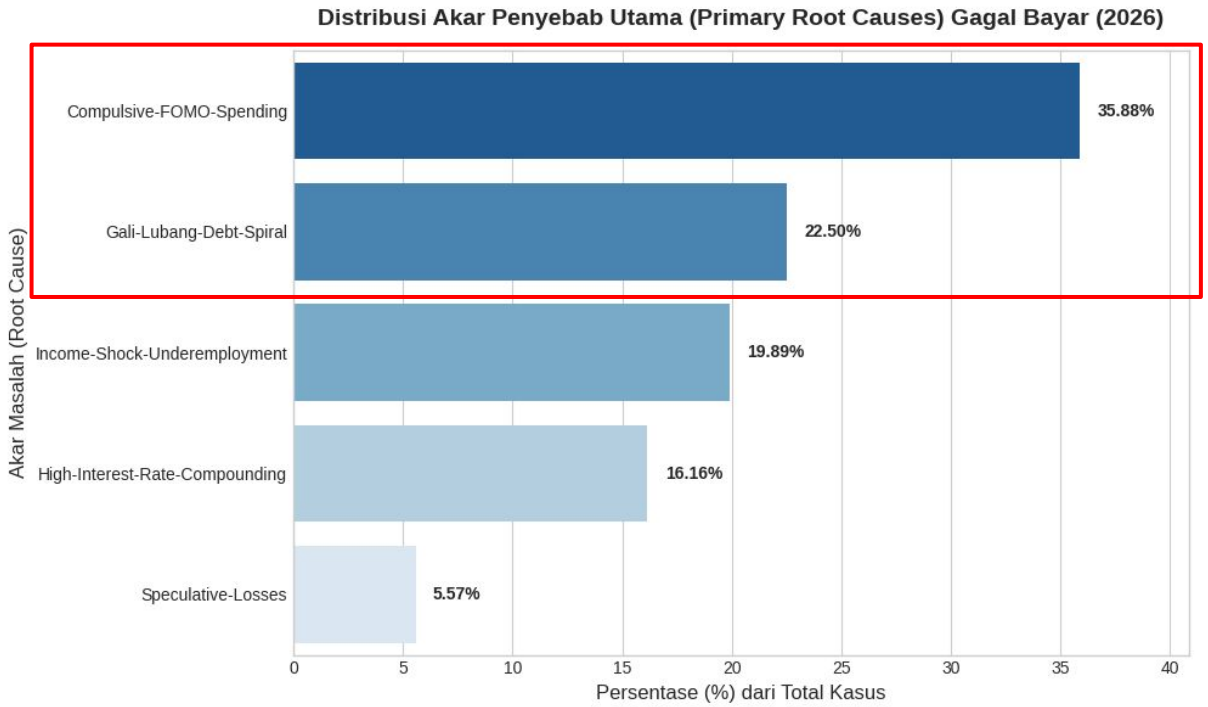
2. Maxed Limit as Crisis Indicator

Limit utilization approaching 87%–100% signals borrowers using paylater products for emergency survival (cash-out distress).

3. DPD 1–30 Escalation

Special Mention borrowers already exhibit a 0.53× DSR. Without early intervention, DSR rapidly accelerates toward the 1.68× tipping point.

Impulsive Spending (FOMO) & Debt Spirals Dominate 58% of Cases



35.88% Lifestyle Spending (FOMO)

Digital credit is predominantly used for gadget upgrades, concert tickets, and non-essential lifestyle goods beyond income capacity

22.50% Debt Spiral Trap

Borrowers take new loans to cover existing debt installments, accelerating personal bankruptcy due to interest compounding.

Secondary External Shocks

Income shock (19.89%) ranks third, indicating lifestyle/character screening improvements are paramount.

Interactive Risk Monitoring Dashboard



Akses Live Dashboard

Buka Tableau Public Interactive Portfolio

Customer Risk Segmentation (Tiering Framework)

TIER 3 (57.15%)

High Risk

Rule: DSR > 0.8 | Limit Util > 80% | Apps ≥ 5

Largest segment, reflecting severe debt spiral exposure.

ACTION STRATEGIS
Auto-reject new loan requests, freeze credit limits, & offer structured debt workout.

TIER 2 (30.59%)

Medium Risk

Rule: DSR 0.51–0.8 | Util 60–80% | Apps 3–4

Transition segment (Special Mention DPD 1–30), experiencing initial financial stress.

ACTION STRATEGIS
Lock automatic limit upgrades, trigger automated reminders, & deliver financial literacy modules.

TIER 1 (12.26%)

Low Risk

Rule: DSR ≤ 0.5 | Util < 60% | Apps ≤ 2

Healthiest borrowers with safe cash flow and controlled debt ratios.

ACTION STRATEGIS
Maintain prime privileges, loyalty rewards, & periodic limit upsell opportunities.

Rekomendasi Bisnis & Action Plan Roadmap

Timeline	Business Focus	Tactical Action	Target KPI
Short-Term (1–4 Weeks)	**Emergency Mitigation & Gate Tightening:** Validate DSR & lock high-risk limits.	Auto-reject apps if DSR > 0.8× or active apps > 3. Freeze limit drawdowns for DSR > 0.5×.	**Default Rate < 20%** (Initial simulation drops to 0.06%).
Medium-Term (1–3 Months)	**Early Warning System & Early Restructuring:** Intervene prior to default.	Trigger automated EWS alerts at DPD 1–30. Offer tenor extensions without punitive penalty compounding.	**Prevent 17.31%** Special Mention segment from deteriorating into hard default.
Long-Term (3–6 Months)	**Credit Scoring Redesign & Literacy:** Underwriting model re-engineering.	Integrate regional UMR baselines & multi-app borrowing metrics into scoring engine. Enforce dynamic limit caps on lifestyle transactions.	**Sustainable Default Rate < 10%**

Thank You

Questions, Feedback & Open Discussion Welcome

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